The second Kahn–Kalai conjecture

Rewritten for readability from the OpenAI manuscript of the same title

Abstract

We prove the second Kahn–Kalai conjecture. For every finite simple graph H with $h \geq 1$ edges and at most n vertices, the threshold for $G(n,p)$ to contain an ordinary copy of H is at most $Cp_{\mathrm{E}}(n,H)(1+\log_2 h)$, where C is universal. Here $p_{\mathrm{E}}(n,H)$ is the least density at which every subgraph of H has expected copy count at least one half.

1 The main result and how it is proved

1.1 Definitions and statement

All graphs are finite and simple; $v(F)$ and $e(F)$ are the numbers of vertices and edges of F. A *subgraph* of G is a graph whose vertex set and edge set are subsets of those of G. A *copy* of F in G is a subgraph of G isomorphic to F, and $N(G,F)$ is the number of copies. These are ordinary copies: G may have additional edges between the vertices of a copy. Let K_n be the complete graph on $[n]=\{1,\dots,n\}$ and let $G(n,p)$ be the random graph on $[n]$ in which each edge of K_n is present independently with probability p. Each copy of F in K_n has $e(F)$ edges and so lies in $G(n,p)$ with probability $p^{e(F)}$; by linearity of expectation,

$$\mathbb{E}N(G(n,p),F) = N(K_n,F)\,p^{e(F)}. \tag{1.1}$$

For a graph H with at least one edge and at most n vertices, define

$$p_{\mathrm{c}}(n,H) = \inf\{p\in[0,1] : \mathbb{P}(N(G(n,p),H)>0) \geq \tfrac{1}{2}\},$$
$$p_{\mathrm{E}}(n,H) = \inf\{p\in[0,1] : \mathbb{E}N(G(n,p),F) \geq \tfrac{1}{2} \text{ for every subgraph } F\subseteq H\}.$$

The first is the *containment threshold*, the second the *graph expectation threshold*. Three elementary facts:

(a) *Both infima are attained.* The probability and the expectations are polynomials in p, hence continuous, so each set of admissible p is closed; there are finitely many constraints; and $p=1$ satisfies all of them, because K_n contains a copy of H (as $v(H)\leq n$) and hence of every $F\subseteq H$.

(b) $p_{\mathrm{E}}(n,H)>0$. A single edge is a subgraph of H, and by (1.1) its constraint is $\binom{n}{2}p \geq \frac{1}{2}$.

(c) $p_{\mathrm{E}}(n,H) \leq p_{\mathrm{c}}(n,H)$. By (a) the defining inequality of p_{c} holds at $p=p_{\mathrm{c}}(n,H)$ itself: there $G(n,p)$ contains H, hence every $F\subseteq H$, with probability at least $\frac{1}{2}$, so $\mathbb{E}N(G(n,p),F) \geq \mathbb{P}(N(G(n,p),F)\geq 1) \geq \frac{1}{2}$.

Theorem 1.1. *For every $n\geq 2$ and every finite simple graph H with $h=e(H)\geq 1$ edges and at most n vertices,*

$$p_{\mathrm{c}}(n,H) \leq \min\{1,\ 2048\mathrm{e}^{50}p_{\mathrm{E}}(n,H)(1+\log_2 h)\}. \tag{1.2}$$

In particular,

$$p_{\mathrm{c}}(n,H) \leq 6144\mathrm{e}^{50}p_{\mathrm{E}}(n,H)\log_2 n.$$

Here e is the base of the natural logarithm (not to be confused with the edge count $e(\cdot)$). The constants are explicit and universal; H may vary with n.

Normalization. Kahn and Kalai use expected count one instead of one half. Write $p_{\mathrm{E}}^{(1)}(n,H)$ for the threshold defined with 1 in place of $\frac{1}{2}$ (this symbol is used only in this remark). Then $p_{\mathrm{E}} \le p_{\mathrm{E}}^{(1)} \le 2p_{\mathrm{E}}$. Indeed, by (1.1), for a subgraph F with $e \ge 1$ edges the constraint $\mathbb{E}N(G(n,p),F) \ge \theta$ reads $p \ge (\theta/N(K_n,F))^{1/e}$, and passing from $\theta = \frac{1}{2}$ to $\theta = 1$ multiplies this lower bound by $2^{1/e} \in (1,2]$; a subgraph without edges has constant expected count $N(K_n,F) \ge 1$ and constrains p in neither case. Each threshold is the largest of these lower bounds. Since p_{c} does not involve the normalization, the two versions of the conjecture differ only in the value of the universal constant.

Conventions. log is the natural logarithm. For a finite set X and $0 \le \rho \le 1$, μ_ρ is the law of a random subset of X that contains each element independently with probability ρ; we also write $\mu_\rho(y) = \rho^{|y|}(1-\rho)^{|X|-|y|}$ for the probability of a set $y \subseteq X$. A graph on $[n]$ is determined by its edge set, a subset of $X = \binom{[n]}{2}$, and the edge set of $G(n,p)$ has law μ_p.

1.2 Proof at a glance

What has to be shown. Let r be the right-hand side of (1.2). By the definition of p_{c} it suffices to show that a μ_r-random subset of $X = \binom{[n]}{2}$ contains the edge set of a copy of H with probability at least $\frac{1}{2}$. The proof gives $\frac{2}{3}$.

Key idea. A copy of H is built in stages $\varnothing = H_0 \subset H_1 \subset \cdots \subset H_k$ (edge sets, H_k being all of H), each stage adding more than 16 times as many edges as the previous one. The stages are chosen so that, given any copy of H_{i-1}, a uniformly random extension to a copy of H_i adds an edge set whose law is *σ-spread* with $\sigma = 128\,p_{\mathrm{E}}(n,H)$: every fixed set J of edges is contained in the added set with probability at most $\sigma^{|J|}$. All stage-by-stage constructions are stored in one rooted tree whose arcs are labelled by the added edge sets. A random set W of density $\rho = 4\mathrm{e}^{50}\sigma$ is then used to shrink *all* arcs of the tree at once: each label is replaced by its part outside W, and enough of the tree survives that the size bound at every level is halved, while the spread constant barely changes. After $\lfloor \log_2 h \rfloor + 1$ such rounds nothing is left to cover, so the union of the sets W that were used contains a copy of H. Using the same W at all levels creates dependence between levels. The mechanism that controls it is this: the labels along any root-to-leaf path of the tree are pairwise disjoint (successive stages add disjoint edge sets), so everything computed in the subtree below an arc depends on W only outside that arc's label (Lemma 3.2, Step 4), and for such quantities Lemma 2.1 gives a change-of-measure bound (Lemma 3.2, Step 5).

The steps and their dependencies.

1. **Lemma 2.1 (one family).** For a single a-spread family of sets of size at most m and $W \sim \mu_\rho$ with $a/\rho \le \mathrm{e}^{-50}$, a random "transition" replaces a set A of the family by a fragment $T = A \setminus W$. The lemma gives (i) $|T| \le m/2$ except with probability e^{-9m}, and (ii) a change-of-measure bound with cost e^{11m} for functions of W outside A.
2. **Lemma 3.2 (one round on a tree).** Applies step 1 at every node with one common W. Part (ii) of step 1 together with disjointness along paths shows that, with probability at least $\frac{3}{4}$, a subtree survives in which all capacities are halved.

3. **Theorem 3.1 (tree covering).** Iterates step 2. This is where the factor $1 + \log_2 \ell_k$ arises: it is the number of halvings.
4. **Lemma 4.1 (graph hierarchy).** Builds the tree for H. This is the only place where graphs and p_{E} enter.
5. **Theorem 1.1** is Theorem 3.1 applied to the tree of Lemma 4.1.

$$\text{Lemma 2.1} \Longrightarrow \text{Lemma 3.2} \Longrightarrow \text{Theorem 3.1} \Longrightarrow \text{Theorem 1.1} \Longleftarrow \text{Lemma 4.1}.$$

Sections 2 and 3 concern set systems only and can be read without Section 4.

Where the constant comes from. $2048 = 128 \cdot 4 \cdot 4$: the hierarchy has spread $\sigma = 128\, p_{\mathrm{E}}$; over all rounds the spread constant grows by a factor less than 4; and $4s$ rounds are allowed for $s = \lfloor \log_2 \ell_k \rfloor + 1$ successes. The factor e^{50} is the ratio ρ/a required by Lemma 2.1.

Notation changes relative to the source. Only names are changed, to give each symbol one meaning. The random source and target of the transition are written c, b, and c, b denote their values. The summation index in Z is b'. In Lemma 3.2 the outgoing level of the current node is always j (the source also used s), and summation indices are i, g (the source used j, h). In the proof of Theorem 3.1 the number of successes so far is τ (source: t). In Section 4 the number $p_{\mathrm{E}}(n, H)$ is written q_* (source: q), to keep it apart from the functionals $q(\cdot)$, $q_f(\cdot)$ of Section 1.3; a uniform random copy of I is $\mathbf{\Gamma}_I$ (source: $\boldsymbol{I}$), $|S|$ is κ (source: r), and the edge set of H is written $E(H)$ rather than H.

1.3 Context and attribution (not used in the proofs)

Nothing in this subsection is used in Sections 2–4. The statements quoted here are results of the cited works; they are reported, not proved or rechecked, here.

The conjecture. By fact (c), a necessary condition for containing H is that every subgraph of H have a reasonable expected number of copies. Kahn and Kalai [8] conjectured that this obstruction determines the containment threshold within a logarithmic factor, uniformly over the target graph. Theorem 1.1 proves this, with the logarithm depending on the number of target edges. For a fixed target, threshold questions belong to the classical theory of Erdős and Rényi [5]; Bollobás [2] treated general fixed subgraphs. A logarithm is needed when the target grows: for H a perfect matching on an even number n of vertices, $p_{\mathrm{E}}(n, H) = \Theta(1/n)$ whereas $p_{\mathrm{c}}(n, H) = \Theta(\log n/n)$ [6, 8], the larger scale being needed to eliminate isolated vertices of the host. Hamilton cycles exhibit the same gap [8]. Other spanning targets have required different methods: Riordan [14], for example, proved that for every fixed $p > 1/4$, $G(2^d, p)$ contains the d-dimensional cube with probability tending to one.

The abstract conjecture, and why it does not suffice. Let $\mathcal{F} \subseteq 2^X$ be a nonempty proper increasing family on a finite set X. A *cover* is a family $\mathcal{G} \subseteq 2^X$ such that every $F \in \mathcal{F}$ contains some $S \in \mathcal{G}$; its cost at density p is $\sum_{S \in \mathcal{G}} p^{|S|}$. The integral expectation threshold $q(\mathcal{F})$ is the largest p for which some cover has cost at most $\frac{1}{2}$. A fractional cover assigns weights $w_S \geq 0$ with $\sum_{S \subseteq F} w_S \geq 1$ for each $F \in \mathcal{F}$, has cost $\sum_S w_S p^{|S|}$, and defines $q_f(\mathcal{F})$ with the same budget. Union bounds and expectations give $q(\mathcal{F}) \leq q_f(\mathcal{F}) \leq p_{\mathrm{c}}(\mathcal{F})$, where $p_{\mathrm{c}}(\mathcal{F})$ is the density at which a μ_p-random subset belongs to $\mathcal{F}$ with probability $\frac{1}{2}$. Talagrand [15, Section 6] introduced the

fractional formulation and its dual description by spread probability measures. Alweiss, Lovett, Wu, and Zhang [1], building on the regularity viewpoint of Lovett, Solomon, and Zhang [9], developed the fragment method in their work on sunflowers. Frankston, Kahn, Narayanan, and Park [7] proved $p_{\mathrm{c}}(\mathcal{F}) = O(q_f(\mathcal{F}) \log \ell)$, where ℓ is the maximum of two and the largest size of a minimal member of $\mathcal{F}$ (this ℓ is unrelated to the capacities ℓ_i of Section 3). Park and Pham [13] proved the corresponding bound with q in place of q_f, resolving the abstract Kahn–Kalai conjecture by a minimum-fragment argument. The integral and fractional thresholds are within a universal constant factor at the common one-half budget [12, Theorem 1.1]. This comparison is contextual; the proof below does not use it.

Let $\mathcal{F}_H$ be the family of edge sets on $[n]$ containing a copy of H. Then $p_{\mathrm{E}}(n, H) \le q(\mathcal{F}_H)$, for the following reason. Take any $p < p_{\mathrm{E}}(n, H)$ (possible by fact (b)). By the definition of p_{E} some subgraph $F \subseteq H$, depending on p, has $\mathbb{E}N(G(n,p), F) < \frac{1}{2}$. Every member of $\mathcal{F}_H$ contains a copy of H, hence the edge set of a copy of F. So the edge sets of the copies of F in K_n form a cover, and by (1.1) its cost at p is at most $\mathbb{E}N(G(n,p), F) < \frac{1}{2}$. Hence $q(\mathcal{F}_H) \ge p$ for every $p < p_{\mathrm{E}}(n, H)$. The inequality goes the wrong way for our purpose: the abstract theorem bounds p_{c} by $O(q(\mathcal{F}_H) \log)$, which does not bound it in terms of the possibly smaller $p_{\mathrm{E}}(n, H)$.

Earlier graph results. Mossel, Niles-Weed, Sun, and Zadik [11, Theorem 1] obtained a single logarithm using the modified graph threshold

$$\widetilde{p}_{\mathrm{E}}(n, H) = \min\{p : \mathbb{E}N(G(n,p), F) \ge N(H, F)/2 \text{ for every } F \subseteq H\},$$

where the factor $N(H, F)$ records that containing H supplies all its copies of F. By symmetry this parameter equals $q_f(\mathcal{F}_H)$ [3, Equation (1)]; this identity is quoted from that reference and is not proved or used here. Comparing it with $p_{\mathrm{E}}(n, H)$ is the additional graph-specific problem. Dubroff, Kahn, and Park [3, Theorems 1.2–1.5] proved $q_f(\mathcal{F}_H) = O(p_{\mathrm{E}}(n, H) \log^2 n)$ and hence $p_{\mathrm{c}}(n, H) = O(p_{\mathrm{E}}(n, H) \log^3 n)$, using a rooted-extension decomposition, and established the single-logarithm bound when $p_{\mathrm{E}}(n, H) < 1/(3n)$. In further work [4, Theorems 1.4–1.5] they proved $N(H, F) \le \mathbb{E}N(G(n, \min\{1, Lq\}), F)$ whenever all subgraphs of H have expected count at least one at a density q, for test graphs F that are cliques, cycles, or bounded-degree trees (L universal for cliques and cycles, depending on the degree bound for trees). Tran [16] then obtained $q_f(\mathcal{F}_H) = O(p_{\mathrm{E}}(n, H) \log(2e(H)))$, reducing the general containment loss to $O(\log^2(2e(H)))$, and removed the loss in the comparison of expectation thresholds for trees and for graphs whose average degree is at least the logarithm of their maximum degree, thereby proving the second conjecture for these classes.

What is taken from earlier work and what is new. In the account above, the route through $q_f(\mathcal{F}_H)$ pays two logarithms: one in bounding $q_f(\mathcal{F}_H)$ by p_{E}, and one in passing from q_f to p_{c} via [7]. The present proof never forms $q_f(\mathcal{F}_H)$, that is, it never needs one spread measure on whole copies of H. From Tran [16, Section 3, Lemma 3.1] it takes the ratio-maximizing choice of a subgraph $S \subseteq I$ and the resulting spread of each *conditional* extension law; this costs the constant 128 and no logarithm (Lemma 4.1, proved in full below). The conditioning step has antecedents in the large-link argument of Lovett, Solomon, and Zhang [9, Claim 29] and the spread-link extraction from a maximal violating core in Alweiss, Lovett, Wu, and Zhang [1, Lemma 2.4]. The copy-count identity (4.1) also appears in [11, Equation (5)]. The transition of Section 2 is the planted-set and posterior-resampling construction of Mossel, Niles-Weed, Sun, and Zadik [10, Sections 2.1–2.3]; Lemma 2.1 adds a capacity-dependent failure estimate and a change-of-measure bound. Tran [16, Section 4] proposed a coupling conjecture that would absorb successive conditional extensions

while paying the logarithmic sampling cost once. Theorem 3.1 proves the required containment bound instead by a tree reduction: all possible extension histories are kept in a tree, every level is reduced with the same random set, and the logarithm is paid once.

2 Resampling one spread family

This section concerns a single weighted family of sets. Its result, Lemma 2.1, is applied in Section 3 at every node of a tree.

Weighted families and spread. Throughout this section X is a finite set and $0 < \rho < 1$. A *weighted family* on X consists of a finite index set B, a *label* $A_b \subseteq X$ for each $b \in B$, and weights $\nu_b > 0$ with $\sum_{b\in B} \nu_b = 1$. Distinct indices may have the same label, and labels may be empty. For $a > 0$ the family is *a-spread* if

$$\mathbb{P}_{c\sim\nu}(J \subseteq A_c) = \sum_{b\in B:\, J\subseteq A_b} \nu_b \le a^{|J|} \qquad (\varnothing \neq J \subseteq X). \tag{2.1}$$

The transition. For $y \subseteq X$ put

$$Z(y) = \sum_{b'\in B} \nu_{b'} \rho^{-|A_{b'}|} \mathbf{1}_{\{A_{b'}\subseteq y\}}. \tag{2.2}$$

Given a set $W \subseteq X$, the *transition at W* produces a random pair $(\mathsf{c}, \mathsf{b}) \in B \times B$ as follows.

1. Choose a *source* c with law ν, and set $Y = W \cup A_{\mathsf{c}}$.
2. Given $\mathsf{c} = c$, choose a *target* b with

$$\mathbb{P}(\mathsf{b} = b \mid \mathsf{c} = c) = \pi_W(b \mid c) := \frac{\nu_b \rho^{-|A_b|} \mathbf{1}_{\{A_b\subseteq W\cup A_c\}}}{Z(W \cup A_c)}. \tag{2.3}$$

3. Retain the *fragment* $T = A_{\mathsf{b}} \setminus W$.

This is well defined: $Z(W \cup A_c) \ge \nu_c \rho^{-|A_c|} > 0$, and $\sum_b \pi_W(b \mid c) = 1$ by (2.2). So at a fixed W the joint law is

$$\mathbb{P}(\mathsf{c} = c,\ \mathsf{b} = b \mid W) = \nu_c\, \pi_W(b \mid c). \tag{2.4}$$

Two regimes. "At fixed W" means that W is a given set and probabilities are over (c, b) with law (2.4). "With $W \sim \mu_\rho$" means that W is drawn first and then (c, b) is drawn by (2.4), so the triple $(W, \mathsf{c}, \mathsf{b})$ has law $\mu_\rho(W)\nu_c\pi_W(b \mid c)$. In both regimes the source has law ν, and in the second it is independent of W, since the c-marginal of (2.4) is ν_c for every W. The target is different: at fixed W its law is $\sum_c \nu_c \pi_W(b \mid c)$, which in general is not ν_b.

Four properties of the transition.

(P1) *The fragment lies in the source label:* $T \subseteq A_{\mathsf{c}} \cap A_{\mathsf{b}}$. Indeed $\pi_W(b \mid c) > 0$ forces $A_b \subseteq W \cup A_c$, so $A_b \setminus W \subseteq A_c$. This is used to transfer spread from the source to the fragment.

(P2) *W and the fragment cover the target label:* $A_{\mathsf{b}} \subseteq W \cup T$, by the definition of T. This does not determine A_{b} from W and T; only the covering is used.

(P3) *Locality:* the law (2.4), the value $Z(W \cup A_c)$ for each c, and the set $A_b \setminus W$ for each b depend on W only through $W \cap \bigcup_{b' \in B} A_{b'}$. Indeed $\mathbf{1}_{\{A_{b'} \subseteq W \cup A_c\}} = \mathbf{1}_{\{A_{b'} \setminus A_c \subseteq W\}}$, and these indicators determine Z and π_W.

(P4) *Law of the planted set:* if $W \sim \mu_\rho$, then $\mathbb{P}(Y = y) = \mu_\rho(y) Z(y)$ for every $y \subseteq X$.

Proof of (P4). Fix c. We have $W \cup A_c = y$ if and only if $A_c \subseteq y$, every element of $y \setminus A_c$ lies in W, and no element of $X \setminus y$ lies in W; elements of A_c are unconstrained. For $A_c \subseteq y$ this has probability $\rho^{|y|-|A_c|}(1-\rho)^{|X|-|y|} = \mu_\rho(y)\rho^{-|A_c|}$. Since c is independent of W, averaging over c with weights ν_c gives $\mu_\rho(y)Z(y)$. (Sanity check: summing over y gives $\mathbb{E}_{\mu_\rho} Z = \sum_b \nu_b \rho^{-|A_b|}\rho^{|A_b|} = 1$.) □

Why (2.3) is called a posterior (not needed later). By the proof of (P4), when $W \sim \mu_\rho$ the conditional law of the source given $Y = y$ is $\nu_c \rho^{-|A_c|}\mathbf{1}_{\{A_c \subseteq y\}}/Z(y)$, which is (2.3). So the target is a second sample from the conditional law of the source given Y.

Lemma 2.1 (Local resampling)**.** *Let $(A_b)_{b\in B}$ with weights ν be a weighted family on X and m a positive integer. Suppose that $|A_b| \le m$ for all b, that the family is a-spread, and that*

$$0 < \rho < 1, \qquad a/\rho \le \mathrm{e}^{-50}.$$

Let $W \sim \mu_\rho$ and perform the transition at W. Call the outcome a local failure *if*

$$Z(Y) < \mathrm{e}^{-10m} \quad \text{or} \quad |T| > m/2.$$

Then

$$\mathbb{P}(\text{local failure}) \le \mathrm{e}^{-9m}. \tag{2.5}$$

Moreover, for each fixed $b \in B$ and every nonnegative function f of subsets of X such that $f(W)$ depends only on $W \setminus A_b$,

$$\mathbb{E}\Big[\mathbf{1}_{\{\mathsf{b}=b,\ Z(Y)\ge \mathrm{e}^{-10m}\}} f(W)\Big] \le \mathrm{e}^{11m}\nu_b\, \mathbb{E} f(W). \tag{2.6}$$

On the left of (2.5) and (2.6) the randomness is the triple $(W, \mathsf{c}, \mathsf{b})$; on the right of (2.6) it is $W \sim \mu_\rho$ alone.

Use. In the proof of Lemma 3.2, (2.5) bounds the failure mass at each node and (2.6) is applied with f equal to a quantity computed in the subtree below the target.

Proof. Plan. Step 1 bounds the event $Z(Y) < \mathrm{e}^{-10m}$ using (P4). Where $Z(Y) \ge \mathrm{e}^{-10m}$, the denominator in (2.3) is bounded below, and everything reduces to one computation for a fixed pair (c, b) (Step 3) combined with the spread hypothesis (Step 2).

Step 1 (small denominator). By (P4),

$$\mathbb{P}(Z(Y) < \mathrm{e}^{-10m}) = \sum_{y \subseteq X} \mu_\rho(y) Z(y) \mathbf{1}_{\{Z(y) < \mathrm{e}^{-10m}\}} \le \mathrm{e}^{-10m} \sum_y \mu_\rho(y) = \mathrm{e}^{-10m}. \tag{2.7}$$

Step 2 (spread bounds intersections). Fix $b \in B$ and an integer $0 \le i \le m$. If $|A_c \cap A_b| = i$, then some i-element $J \subseteq A_b$ satisfies $J \subseteq A_c$. By the union bound, (2.1) (trivially true for $i = 0$) and $|A_b| \le m$,

$$\mathbb{P}_{c\sim\nu}(|A_c \cap A_b| = i) \le \sum_{\substack{J \subseteq A_b \\ |J| = i}} \mathbb{P}_{c\sim\nu}(J \subseteq A_c) \le \binom{m}{i} a^i. \tag{2.8}$$

Consequently, for any function $g \ge 0$ on $\{0, \dots, m\}$, grouping the sources c by the value $i = |A_c \cap A_b|$ gives

$$\sum_c \nu_c \, g(|A_c \cap A_b|) = \sum_{i=0}^{m} \mathbb{P}_{c\sim\nu}(|A_c \cap A_b| = i) g(i) \le \sum_{i=0}^{m} \binom{m}{i} a^i g(i). \tag{2.9}$$

Step 3 (one pair). Fix $c, b \in B$ and $f \ge 0$ with $f(W)$ depending only on $W \setminus A_b$. We claim

$$\mathbb{E}_{W\sim\mu_\rho}\Big[\pi_W(b \mid c)\, \mathbf{1}_{\{Z(W\cup A_c)\ge \mathrm{e}^{-10m}\}} f(W)\Big] \le \mathrm{e}^{10m}\, \nu_b\, \rho^{-|A_c\cap A_b|}\, \mathbb{E} f(W). \tag{2.10}$$

On the event in the indicator, $1/Z(W \cup A_c) \le \mathrm{e}^{10m}$, so by (2.3) and $\mathbf{1}_{\{A_b \subseteq W\cup A_c\}} = \mathbf{1}_{\{A_b\setminus A_c \subseteq W\}}$ the integrand is at most $\mathrm{e}^{10m}\nu_b\rho^{-|A_b|}\mathbf{1}_{\{A_b\setminus A_c\subseteq W\}} f(W)$. Under μ_ρ the coordinates are independent. The event $\{A_b \setminus A_c \subseteq W\}$ is determined by $W \cap A_b$ and has probability $\rho^{|A_b\setminus A_c|}$, while $f(W)$ is determined by $W \setminus A_b$. Hence

$$\rho^{-|A_b|}\, \mathbb{E}\big[\mathbf{1}_{\{A_b\setminus A_c\subseteq W\}} f(W)\big] = \rho^{-|A_b|+|A_b\setminus A_c|}\, \mathbb{E} f(W) = \rho^{-|A_c\cap A_b|}\, \mathbb{E} f(W),$$

which gives (2.10).

Step 4 (proof of (2.5)). By (2.4),

$$\mathbb{P}(Z(Y) \ge \mathrm{e}^{-10m},\ |T| > \tfrac{m}{2}) = \sum_{c,b} \nu_c\, \mathbb{E}_W\Big[\pi_W(b \mid c) \mathbf{1}_{\{Z(W\cup A_c)\ge \mathrm{e}^{-10m}\}} \mathbf{1}_{\{|A_b\setminus W|>m/2\}}\Big].$$

When $\pi_W(b \mid c) > 0$, (P1) gives $A_b \setminus W \subseteq A_c \cap A_b$, so the last indicator is at most $\mathbf{1}_{\{|A_c\cap A_b|>m/2\}}$, which does not depend on W. Applying (2.10) with $f \equiv 1$ to each pair, and then (2.9) with $g(i) = \rho^{-i}\mathbf{1}_{\{i>m/2\}}$ for each b,

$$\begin{aligned}
\mathbb{P}(Z(Y) \ge \mathrm{e}^{-10m},\ |T| > \tfrac{m}{2}) &\le \mathrm{e}^{10m} \sum_b \nu_b \sum_c \nu_c \rho^{-|A_c\cap A_b|} \mathbf{1}_{\{|A_c\cap A_b|>m/2\}} \\
&\le \mathrm{e}^{10m} \sum_b \nu_b \sum_{m/2<i\le m} \binom{m}{i} (a/\rho)^i \\
&\le \mathrm{e}^{10m} \cdot \mathrm{e}^{-25m} \sum_{i=0}^{m} \binom{m}{i} = \mathrm{e}^{10m}\, \mathrm{e}^{-25m}\, 2^m = \mathrm{e}^{-(15-\log 2)m}.
\end{aligned}$$

The third line uses $(a/\rho)^i \le \mathrm{e}^{-50i} < \mathrm{e}^{-25m}$ for $i > m/2$ and $\sum_b \nu_b = 1$. With (2.7),

$$\mathbb{P}(\text{local failure}) \le \mathrm{e}^{-10m} + \mathrm{e}^{-(15-\log 2)m} = \mathrm{e}^{-9m}\big(\mathrm{e}^{-m} + \mathrm{e}^{-(6-\log 2)m}\big) \le \mathrm{e}^{-9m},$$

because for $m \ge 1$ the bracket is at most $\mathrm{e}^{-1} + 2\mathrm{e}^{-6} < 0.38 < 1$.

Step 5 (proof of (2.6)). Fix b and f. By (2.4), then (2.10), then (2.9) with $g(i) = \rho^{-i}$,

$$\begin{aligned}
\mathbb{E}\Big[\mathbf{1}_{\{\mathsf{b}=b,\ Z(Y)\ge \mathrm{e}^{-10m}\}} f(W)\Big] &= \sum_c \nu_c\, \mathbb{E}_W\Big[\pi_W(b \mid c) \mathbf{1}_{\{Z(W\cup A_c)\ge \mathrm{e}^{-10m}\}} f(W)\Big] \\
&\le \mathrm{e}^{10m} \nu_b\, \mathbb{E} f(W) \sum_c \nu_c \rho^{-|A_c\cap A_b|} \\
&\le \mathrm{e}^{10m} \nu_b\, \mathbb{E} f(W) \sum_{i=0}^{m} \binom{m}{i} (a/\rho)^i = \mathrm{e}^{10m} \nu_b\, \mathbb{E} f(W)\, (1 + a/\rho)^m.
\end{aligned}$$

Finally $(1 + a/\rho)^m \le (1 + \mathrm{e}^{-50})^m \le \mathrm{e}^m$. □

3 Covering a tree

The input is a finite tree whose arc labels are disjoint along each path and whose child laws are spread. The aim is to put all labels of one path inside a product sample, at a cost logarithmic in the largest label size.

Definitions. A *probability tree on X* is a finite rooted tree in which all leaves have the same depth $k \geq 0$, together with, for each non-leaf node v, a probability distribution ν_v with all weights positive on the set $\mathrm{Ch}(v)$ of children of v, and a label $A_v(b) \subseteq X$ on the arc from v to each child b. Different arcs may carry the same label; nodes are never identified because labels agree.

- *Levels.* For $1 \leq j \leq k$, *level j* is the set of arcs from depth $j-1$ to depth j. A node at depth $j-1$ has *outgoing level j*: the root has outgoing level 1, the parents of leaves have outgoing level k, and leaves have none. The tree has k levels; if $k = 0$ it is a single node.
- *Paths.* A *path* always means a full root-to-leaf path P, and $\mathcal{U}(P) = \bigcup_{(v,b)\in P} A_v(b)$ is the union of its labels. (For $k = 0$ the only path has $\mathcal{U}(P) = \varnothing$.)
- The tree is *disjoint* if the labels along each path are pairwise disjoint.
- Level j has *capacity m* if all its labels have size at most m.
- Level j is *a-spread* if for every node v with outgoing level j the weighted family $(A_v(b))_{b\in\mathrm{Ch}(v)}$ with weights ν_v is a-spread in the sense of (2.1).

All requirements concern individual child distributions; no spread assumption is made on the distribution of a full path.

Theorem 3.1 (Tree covering)**.** *Let $\mathcal{T}$ be a disjoint probability tree on a finite set X, with $k \geq 1$ levels. Suppose that every level is σ-spread, where $\sigma > 0$, and that the levels have positive integer capacities $\ell_1, \dots, \ell_k$ satisfying*

$$\ell_{i+1} \geq 16\ell_i \qquad (1 \leq i < k).$$

Set

$$r = \min\{1,\ 16\mathrm{e}^{50}\sigma(1 + \log_2 \ell_k)\}.$$

Then

$$\mathbb{P}_{W\sim\mu_r}(\mathcal{U}(P) \subseteq W \text{ for some path } P \text{ of } \mathcal{T}) \geq \tfrac{2}{3}.$$

Use. Section 4 constructs such a tree for H with $\sigma = 128p_{\mathrm{E}}(n, H)$ and $\ell_k \leq e(H)$. *Sanity check:* for $k = 1$ the tree is a single σ-spread family of sets of size at most ℓ_1, and the theorem says that a random set of density $\min\{1, 16\mathrm{e}^{50}\sigma(1 + \log_2 \ell_1)\}$ contains one of them with probability at least $\frac{2}{3}$. The proof repeatedly prunes and reweights the tree while replacing labels by smaller subsets; this is why empty labels and unequal label sizes are allowed in the definitions.

3.1 One simultaneous reduction

Lemma 3.2 (Reduction)**.** *Let $\mathcal{T}$ be a disjoint probability tree on X with $k \geq 1$ levels and positive integer capacities $m_1, \dots, m_k$ satisfying $m_{j+1} \geq 16m_j$. Let $a > 0$ be a real number such that*

$$\rho := \mathrm{e}^{50}a < 1,$$

and suppose that level j is α_j-spread, where $0 < \alpha_j \le a$ $(1 \le j \le k)$. There is a deterministic procedure (whether it succeeds, and what it returns, are functions of $\mathcal{T}$ and of a set $W \subseteq X$) which, for $W \sim \mu_\rho$, succeeds with probability at least $3/4$ over the choice of W, and on success returns a disjoint probability tree $\mathcal{T}'$ with the same levels, such that:

(i) level j has capacity $\lfloor m_j/2 \rfloor$ and is $\alpha_j/(1-\mathrm{e}^{-m_j})$-spread;

(ii) for every path Q of $\mathcal{T}'$ there is a path P of $\mathcal{T}$ with

$$\mathcal{U}(P) \subseteq W \cup \mathcal{U}(Q). \tag{3.1}$$

"The same levels" means that $\mathcal{T}'$ again has k levels (it is a subtree of $\mathcal{T}$ with the same root, new labels and new weights). The lemma deletes no level, even when $\lfloor m_j/2 \rfloor = 0$, in which case all labels of level j of $\mathcal{T}'$ are empty. *Use:* this lemma is iterated in the proof of Theorem 3.1.

Proof. Plan. Run the transition of Section 2 at every node of $\mathcal{T}$ with the same W (Step 1). Call a node good if most of its transition mass leads, without local failure, to good children (Step 2). A potential D_v exceeds 1 at every bad node (Step 3), and its expectation is small (Steps 4–6), so the root is good with probability at least $\frac{3}{4}$. The good nodes reachable from the root form $\mathcal{T}'$ (Steps 7–8). Throughout Steps 1–6 all quantities refer to the fixed input tree $\mathcal{T}$, before any pruning.

Step 1 (local masses). Let v be a node with outgoing level j. The family $(A_v(b))_{b\in\mathrm{Ch}(v)}$ with weights ν_v satisfies the hypotheses of Lemma 2.1 with $m = m_j$ and with α_j in the role of a: its labels have size at most m_j, it is α_j-spread, $0 < \rho < 1$, and $\alpha_j/\rho \le a/\rho = \mathrm{e}^{-50}$. Consider the transition at W for this family, with "local failure" defined using m_j. At fixed W define

$$d_v(W) = \mathbb{P}(\text{local failure at } v \mid W), \qquad t_{vb}(W) = \mathbb{P}(\text{no local failure at } v,\ \mathsf{b} = b \mid W) \quad (b \in \mathrm{Ch}(v)).$$

These are probabilities over the source and target at v only, and

$$d_v(W) + \sum_{b\in\mathrm{Ch}(v)} t_{vb}(W) = 1. \tag{3.2}$$

We record four facts.

(M1) $\mathbb{E}_{W\sim\mu_\rho} d_v(W) \le \mathrm{e}^{-9m_j}$, by (2.5).

(M2) If $f \ge 0$ and $f(W)$ depends only on $W \setminus A_v(b)$, then $\mathbb{E}[t_{vb}(W)f(W)] \le \mathrm{e}^{11m_j}\nu_v(b)\mathbb{E}f(W)$. Indeed "no local failure" includes $Z(Y) \ge \mathrm{e}^{-10m_j}$, so $t_{vb}(W) \le \mathbb{P}(\mathsf{b} = b,\ Z(Y) \ge \mathrm{e}^{-10m_j} \mid W)$; multiply by $f(W)$, average over W, and apply (2.6).

(M3) $d_v(W)$ and all $t_{vb}(W)$ depend on W only through $W \cap \bigcup_{b\in\mathrm{Ch}(v)} A_v(b)$, by (P3).

(M4) If $t_{vb}(W) > 0$, then $|A_v(b) \setminus W| \le \lfloor m_j/2 \rfloor$. Indeed, $t_{vb}(W) > 0$ means that at this W some outcome $(\mathsf{c}, \mathsf{b}) = (c, b)$ with target b is not a local failure. For every outcome with target b the fragment is $T = A_v(b) \setminus W$, whatever the source. "Not a local failure" includes $|T| \le m_j/2$, and $|T|$ is an integer.

Step 2 (good nodes and success). Fix W and put $\epsilon_j = \mathrm{e}^{-m_j}$. Declare all leaves good. Working upward, declare a node v with outgoing level j *good* if

$$\sum_{\substack{b\in\mathrm{Ch}(v)\\ b \text{ good}}} t_{vb}(W) \ge 1 - \epsilon_j, \tag{3.3}$$

and *bad* otherwise. The attempt *succeeds* exactly when the root is good. Splitting the sum in (3.2) into good and bad children gives

$$d_v(W) + \sum_{b \text{ bad}} t_{vb}(W) = 1 - \sum_{b \text{ good}} t_{vb}(W), \tag{3.4}$$

so (3.3) says that the discarded mass at v (failure mass plus mass of bad targets) is at most ϵ_j.

Step 3 (a potential that detects bad nodes). Set $D_v(W) = 0$ at leaves and, at a node v with outgoing level j,

$$D_v(W) = \frac{d_v(W) + \sum_{b \in \mathrm{Ch}(v)} t_{vb}(W) D_b(W)}{\epsilon_j} \geq 0. \tag{3.5}$$

Claim: every bad node has $D_v(W) > 1$. We argue by induction from the leaves upward; leaves are good, so there is nothing to prove for them. Let v be bad with outgoing level j and assume the claim for its children. Using $D_b \geq 0$ for good children and $D_b > 1$ for bad children, the numerator of (3.5) is at least

$$d_v(W) + \sum_{b \text{ bad}} t_{vb}(W) = 1 - \sum_{b \text{ good}} t_{vb}(W) > \epsilon_j,$$

by (3.4) and the failure of (3.3). Hence $D_v(W) > 1$.

Step 4 (D_b ignores W on the arc into b). For a node w let $L_w \subseteq X$ be the union of the labels of all arcs in the subtree below w ($L_w = \varnothing$ for a leaf).

(a) $D_w(W)$ depends on W only through $W \cap L_w$. Induction from the leaves upward: by (3.5), D_w is a function of d_w and the $t_{wb'}$, which depend on W only through $W \cap \bigcup_{b'} A_w(b')$ by (M3), and of the $D_{b'}$ for $b' \in \mathrm{Ch}(w)$, which depend only on $W \cap L_{b'}$ by induction. All these sets lie in L_w.

(b) If b is a child of v, then $L_b \cap A_v(b) = \varnothing$. Any arc below b lies on a root-to-leaf path that passes through the arc (v, b): go from the root to v, then to b, down to the given arc, and on to a leaf. Since $\mathcal{T}$ is disjoint, the label of that arc is disjoint from $A_v(b)$.

By (a) and (b), $D_b(W)$ depends only on $W \setminus A_v(b)$. See Figure 1. This holds for every node of the fixed input tree, with no conditioning on being good or retained.

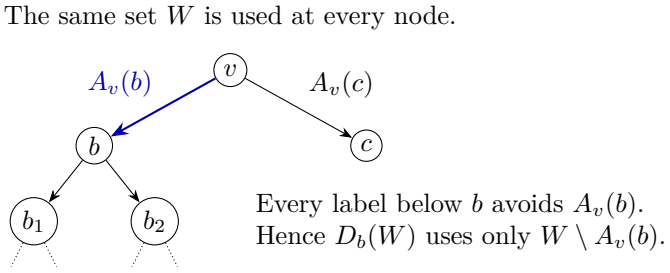


Figure 1: Every arc below b lies on a path through the arc into b, so the potential D_b is a function of coordinates outside $A_v(b)$. The diagram is schematic; labels on different branches (for instance $A_v(c)$ and labels below b) may overlap.

Step 5 (expected potential: one level). Now let $W \sim \mu_\rho$. Let v have outgoing level j. By Step 4 we may apply (M2) with $f = D_b$ for each child b; with (M1) and $\epsilon_j^{-1} = \mathrm{e}^{m_j}$, taking

expectations in (3.5) gives

$$\mathbb{E}D_v(W) \le \mathrm{e}^{m_j}\Big(\mathrm{e}^{-9m_j} + \mathrm{e}^{11m_j}\sum_b \nu_v(b)\,\mathbb{E}D_b(W)\Big) = \mathrm{e}^{-8m_j} + \mathrm{e}^{12m_j}\sum_{b\in\mathrm{Ch}(v)} \nu_v(b)\,\mathbb{E}D_b(W). \quad (3.6)$$

This is the step that controls the dependence caused by reusing W: the transition at v and the potential D_b both depend on W, and (2.6) decouples them at the cost e^{11m_j}.

Step 6 (solving the recursion). For $1 \le j \le k$ let β_j be the maximum of $\mathbb{E}D_v(W)$ over nodes v with outgoing level j, and put $\beta_{k+1} = 0$ (leaves). The children of a node with outgoing level j have outgoing level $j+1$ (or are leaves if $j = k$), and the weights $\nu_v(b)$ sum to 1, so (3.6) gives $\beta_j \le \mathrm{e}^{-8m_j} + \mathrm{e}^{12m_j}\beta_{j+1}$. We claim that for $1 \le j \le k+1$

$$\beta_j \le \sum_{i=j}^{k} \exp\Big(12\sum_{g=j}^{i-1} m_g - 8m_i\Big), \quad (3.7)$$

by downward induction on j. For $j = k+1$ both sides are 0. If (3.7) holds for $j+1$, then

$$\beta_j \le \mathrm{e}^{-8m_j} + \mathrm{e}^{12m_j}\sum_{i=j+1}^{k} \exp\Big(12\sum_{g=j+1}^{i-1} m_g - 8m_i\Big) = \mathrm{e}^{-8m_j} + \sum_{i=j+1}^{k} \exp\Big(12\sum_{g=j}^{i-1} m_g - 8m_i\Big),$$

and e^{-8m_j} is the term $i = j$ of (3.7) (its inner sum is empty). Now take $j = 1$. Since $m_g \le 16^{-(i-g)}m_i$, we have $\sum_{g<i} m_g \le m_i\sum_{x\ge1} 16^{-x} = m_i/15$, so each exponent in (3.7) is at most $(\frac{12}{15} - 8)m_i \le -7m_i$. The m_i are distinct positive integers because $m_{i+1} \ge 16m_i > m_i$. Let o be the root. By Step 3 and Markov's inequality,

$$\mathbb{P}(o \text{ is bad}) \le \mathbb{P}(D_o(W) > 1) \le \mathbb{E}D_o(W) \le \beta_1 \le \sum_{i=1}^{k} \mathrm{e}^{-7m_i} \le \sum_{x\ge1} \mathrm{e}^{-7x} = \frac{1}{\mathrm{e}^7 - 1} < \frac{1}{4}.$$

So the attempt succeeds with probability at least $\frac{3}{4}$.

Step 7 (the reduced tree). Fix W for which the root is good. Start from the root. At any good node v with outgoing level j put

$$\gamma_v(W) = \sum_{b \text{ good}} t_{vb}(W) \ge 1 - \epsilon_j > 0.$$

Retain each good child b with $t_{vb}(W) > 0$, give it weight $t_{vb}(W)/\gamma_v(W)$, replace its incoming label by

$$T_b = A_v(b) \setminus W,$$

and continue recursively into b. Call the result $\mathcal{T}'$. Every retained node is good. The quantities $t_{vb}(W)$ define the construction deterministically; sources and targets are auxiliary random variables used only to define and analyse these numbers.

Step 8 (verification of the conclusions). *Probability tree with k levels.* The new weights at a retained node are positive and sum to 1. Since $\gamma_v > 0$, every retained non-leaf node has at least one retained child, so all leaves of $\mathcal{T}'$ are leaves of $\mathcal{T}$, at depth k.

Disjoint. Paths of $\mathcal{T}'$ are paths of $\mathcal{T}$ and $T_b \subseteq A_v(b)$.

Capacity. A retained arc has $t_{vb}(W) > 0$, so $|T_b| \le \lfloor m_j/2 \rfloor$ by (M4).

Recovery (3.1). A path Q of $\mathcal{T}'$ is, as a sequence of nodes, a path P of $\mathcal{T}$, and $A_v(b) \subseteq W \cup T_b$ on each of its arcs. Taking unions gives $\mathcal{U}(P) \subseteq W \cup \mathcal{U}(Q)$.

Spread. Let v be a retained node with outgoing level j and $\varnothing \neq J \subseteq X$. Under the new child law the probability that the label contains J is

$$\frac{1}{\gamma_v(W)} \sum_{\substack{b \text{ good} \\ J \subseteq A_v(b) \setminus W}} t_{vb}(W).$$

Consider the transition at v at this fixed W. Since $t_{vb}(W) \leq \mathbb{P}(\mathsf{b} = b \mid W)$, and on $\{\mathsf{b} = b\}$ the fragment is $T = A_v(b) \setminus W$, the sum is at most $\mathbb{P}(J \subseteq T \mid W)$. By (P1), $T \subseteq A_v(\mathsf{c})$, and at fixed W the source has law ν_v (see "Two regimes"). Therefore

$$\mathbb{P}(J \subseteq T \mid W) \leq \mathbb{P}_{c \sim \nu_v}(J \subseteq A_v(c)) \leq \alpha_j^{|J|},$$

since level j of $\mathcal{T}$ is α_j-spread. Dividing by $\gamma_v(W) \geq 1 - \epsilon_j$ and using $|J| \geq 1$,

$$\frac{\alpha_j^{|J|}}{1 - \epsilon_j} \leq \left(\frac{\alpha_j}{1 - \epsilon_j} \right)^{|J|}, \qquad \epsilon_j = \mathrm{e}^{-m_j}.$$

This proves (i) and (ii). □

3.2 Iteration

Proof of Theorem 3.1. Plan. Apply Lemma 3.2 repeatedly with fresh independent sets. Each success halves all capacities. After $s = \lfloor \log_2 \ell_k \rfloor + 1$ successes all labels are empty, and the union of the successful sets contains a path union of $\mathcal{T}$. With probability at least $\frac{2}{3}$ this takes at most $4s$ attempts, and the union of $4s$ sets of density ρ has density at most r.

Step 0 (parameters). If $r = 1$ then $W = X$ almost surely and the conclusion is immediate. Suppose $r < 1$, so that $r = 16\mathrm{e}^{50}\sigma(1 + \log_2 \ell_k)$, and set

$$a = 4\sigma, \qquad \rho = \mathrm{e}^{50} a = 4\mathrm{e}^{50}\sigma, \qquad s = \lfloor \log_2 \ell_k \rfloor + 1.$$

Since $1 + \log_2 \ell_k \geq 1$, we get $\rho \leq r/4 < 1$, so $0 < \rho < 1$.

Step 1 (arithmetic of capacities and spread). For integers $\tau \geq 0$ and $1 \leq i \leq k$ define

$$m_i(\tau) = \left\lfloor \frac{\ell_i}{2^\tau} \right\rfloor, \tag{3.8}$$

which is nonincreasing in τ. For the pairs (i, τ) with $\tau = 0$ or $m_i(\tau - 1) \geq 1$ define

$$\alpha_i(\tau) = \sigma \prod_{u=0}^{\tau-1} \left(1 - \mathrm{e}^{-m_i(u)}\right)^{-1}. \tag{3.9}$$

For such a pair, $m_i(u) \geq m_i(\tau - 1) \geq 1$ for all $0 \leq u \leq \tau - 1$, so every factor is finite and positive. These pairs include every (i, τ) with $m_i(\tau) \geq 1$, that is, every level still present after τ successes; $\alpha_i(\tau)$ is not needed for other pairs, except for a level in the round in which its capacity drops to 0.

(A1) $\lfloor m_i(\tau)/2 \rfloor = m_i(\tau + 1)$, because $\lfloor \lfloor x \rfloor / 2 \rfloor = \lfloor x/2 \rfloor$ for real $x \geq 0$ (an integer z satisfies $z \leq x/2$ iff $2z \leq x$ iff $2z \leq \lfloor x \rfloor$).

(A2) $m_i(\tau)$ is nondecreasing in i, so for each τ the levels with $m_i(\tau) = 0$ form an initial segment $i = 1, \dots, k_0$. Moreover $m_{i+1}(\tau) \geq 16 m_i(\tau)$, because $16 m_i(\tau)$ is an integer not exceeding $16\ell_i/2^\tau \leq \ell_{i+1}/2^\tau$.

(A3) $m_k(\tau) \geq 1$ iff $2^\tau \leq \ell_k$ iff $\tau \leq s - 1$. In particular $m_i(s) = 0$ for all i.

(A4) $0 < \alpha_i(\tau) \leq 4\sigma = a$ whenever $\alpha_i(\tau)$ is defined. The integers $m_i(u)$, $0 \leq u \leq \tau - 1$, are at least 1 (as just noted) and strictly decreasing in u: by (A1), $m_i(u+1) = \lfloor m_i(u)/2 \rfloor < m_i(u)$ because $m_i(u) \geq 1$. Hence they are distinct positive integers, and the product in (3.9) is at most $\prod_{x \geq 1} (1 - \mathrm{e}^{-x})^{-1}$. Using $-\log(1-y) \leq y/(1-y)$ for $0 \leq y < 1$ and $\mathrm{e}^{-x} \leq \mathrm{e}^{-1}$,

$$\sum_{x=1}^{\infty} -\log(1 - \mathrm{e}^{-x}) \leq \frac{1}{1 - \mathrm{e}^{-1}} \sum_{x=1}^{\infty} \mathrm{e}^{-x} = \frac{\mathrm{e}^{-1}}{(1 - \mathrm{e}^{-1})^2} < 0.93 < \log 4, \tag{3.10}$$

and exponentiating gives a product less than 4.

Step 2 (the procedure). Let $W_1, W_2, \dots$ be independent sets of law μ_ρ. Maintain a *current tree*, initially $\mathcal{T}$. At attempt $\lambda = 1, 2, \dots$, if the current tree has at least one level, run the procedure of Lemma 3.2 on it with the set W_λ.

- On failure, leave the current tree unchanged.
- On success, let $\mathcal{T}'$ be the returned tree, with k' levels, and *trim* it: let $k_0 \leq k'$ be the number of its levels whose new capacity is 0 (an initial segment, by (A2)); choose by a fixed rule a node o' of $\mathcal{T}'$ at depth k_0; the new current tree is the subtree of $\mathcal{T}'$ rooted at o', with its labels and child laws. Such a node exists because every root-to-leaf path of $\mathcal{T}'$ has a node at each depth $0, \dots, k'$; if $k_0 = k'$ it is a leaf and the new current tree is a single node.

Stop when the current tree has no levels (a single node).

Trimming is harmless. The subtree rooted at o' is a probability tree with $k' - k_0$ levels (the leaves of $\mathcal{T}'$ have depth k'). It is disjoint, since each of its paths is part of a path of $\mathcal{T}'$. Its labels and child laws are those of $\mathcal{T}'$, so every capacity and spread bound of $\mathcal{T}'$ at a retained level still holds. If Q' is a path of the subtree, then the path of $\mathcal{T}'$ from its root to o' followed by Q' is a path Q of $\mathcal{T}'$; its first k_0 labels are empty, so $\mathcal{U}(Q) = \mathcal{U}(Q')$.

Step 3 (invariants). Let $\mathcal{T}_\tau$ be the current tree after τ successes and $\overline{W}_\tau$ the union of the τ sets W_λ of the successful attempts so far ($\overline{W}_0 = \varnothing$). We claim, for every τ reached:

(I1) $\mathcal{T}_\tau$ is a disjoint probability tree whose levels correspond, in order, to the original levels i with $m_i(\tau) \geq 1$, and the level corresponding to i has capacity $m_i(\tau)$;

(I2) that level is $\alpha_i(\tau)$-spread;

(I3) for every path Q of $\mathcal{T}_\tau$ there is a path P of $\mathcal{T}$ with $\mathcal{U}(P) \subseteq \overline{W}_\tau \cup \mathcal{U}(Q)$.

For $\tau = 0$ these are the hypotheses of the theorem. Assume them for τ, with $\mathcal{T}_\tau$ having at least one level, and let W be the set of the next successful attempt. By (I1), (I2), (A2), (A4) and Step 0, $\mathcal{T}_\tau$ satisfies the hypotheses of Lemma 3.2 with capacities $m_i(\tau) \geq 1$, spread parameters $\alpha_i(\tau) \leq a$ and $\rho = \mathrm{e}^{50} a < 1$. Its conclusion (i) and (A1) give capacities $m_i(\tau + 1)$ and spread $\alpha_i(\tau)/(1 - \mathrm{e}^{-m_i(\tau)}) = \alpha_i(\tau + 1)$ for $\mathcal{T}'$. Trimming removes exactly the levels with $m_i(\tau+1) = 0$ and preserves the rest, giving (I1) and (I2) for $\tau + 1$. For (I3): given a path Q' of $\mathcal{T}_{\tau+1}$, trimming gives

a path Q of $\mathcal{T}'$ with $\mathcal{U}(Q) = \mathcal{U}(Q')$; conclusion (ii) gives a path P' of $\mathcal{T}_\tau$ with $\mathcal{U}(P') \subseteq W \cup \mathcal{U}(Q)$; and (I3) for τ gives a path P of $\mathcal{T}$ with

$$\mathcal{U}(P) \subseteq \overline{W}_\tau \cup \mathcal{U}(P') \subseteq \overline{W}_\tau \cup W \cup \mathcal{U}(Q') = \overline{W}_{\tau+1} \cup \mathcal{U}(Q').$$

Only successful draws enter $\overline{W}_\tau$: a failed attempt changes nothing.

Step 4 (completion). By (I1) and (A3), $\mathcal{T}_\tau$ has at least one level for $\tau \le s-1$ and none for $\tau = s$. So the procedure stops exactly at the s-th success. The unique path of the single-node tree $\mathcal{T}_s$ has empty union, so (I3) gives a path P of $\mathcal{T}$ with $\mathcal{U}(P) \subseteq \overline{W}_s$.

Step 5 (number of attempts). *Conditional success probability.* Fix $\lambda \ge 1$ and sets $w_1, \dots, w_{\lambda-1} \subseteq X$ such that, when $W_1 = w_1, \dots, W_{\lambda-1} = w_{\lambda-1}$, the procedure has not stopped before attempt λ. Then the current tree before attempt λ is one fixed tree, a function of $w_1, \dots, w_{\lambda-1}$ (Lemma 3.2 is deterministic and trimming uses a fixed rule); by Step 3 it satisfies the hypotheses of Lemma 3.2; and W_λ is independent of $W_1, \dots, W_{\lambda-1}$ with law μ_ρ. Hence

$$\mathbb{P}(\text{attempt } \lambda \text{ fails} \mid W_1 = w_1, \dots, W_{\lambda-1} = w_{\lambda-1}) \le \tfrac{1}{4}. \tag{3.11}$$

Consequently, if E is an event determined by $W_1, \dots, W_{\lambda-1}$ on which the procedure has not stopped before attempt λ, then

$$\mathbb{P}\big(E \cap \{\text{attempt } \lambda \text{ fails}\}\big) \le \tfrac{1}{4}\,\mathbb{P}(E)\,: \tag{3.12}$$

multiply (3.11) by $\mathbb{P}(W_1 = w_1, \dots, W_{\lambda-1} = w_{\lambda-1})$ and sum over the finitely many tuples that make up E.

Waiting times. For $0 \le i \le s$ let Λ_i be the index of the attempt at which the i-th success occurs ($\Lambda_0 = 0$), and $\Delta_i = \Lambda_i - \Lambda_{i-1}$ for $1 \le i \le s$. We show by induction on i that $\Lambda_{i-1} < \infty$ almost surely and $\mathbb{P}(\Delta_i > x) \le 4^{-x}$ for every integer $x \ge 0$. Assume $\Lambda_{i-1} < \infty$ almost surely (true for $i = 1$). For integers $\lambda, x \ge 0$ let

$$E_{\lambda,x} = \{\Lambda_{i-1} = \lambda \text{ and attempts } \lambda+1, \dots, \lambda+x \text{ all fail}\}.$$

This event is determined by $W_1, \dots, W_{\lambda+x}$, and on it only $i-1 \le s-1$ successes occur among the first $\lambda + x$ attempts, so by Step 4 the procedure has not stopped before attempt $\lambda + x + 1$. Since $E_{\lambda,x+1} = E_{\lambda,x} \cap \{\text{attempt } \lambda + x + 1 \text{ fails}\}$, (3.12) gives $\mathbb{P}(E_{\lambda,x+1}) \le \frac{1}{4}\mathbb{P}(E_{\lambda,x})$. Up to a null set, $\{\Delta_i > x\}$ is the disjoint union of the events $E_{\lambda,x}$ over $\lambda \ge 0$. Summing over λ,

$$\mathbb{P}(\Delta_i > x+1) \le \tfrac{1}{4}\,\mathbb{P}(\Delta_i > x), \qquad \text{hence} \qquad \mathbb{P}(\Delta_i > x) \le 4^{-x}.$$

In particular $\Delta_i < \infty$ and so $\Lambda_i < \infty$ almost surely, which completes the induction.

Markov bound. Therefore

$$\mathbb{E}\Delta_i = \sum_{x \ge 0} \mathbb{P}(\Delta_i > x) \le \sum_{x \ge 0} 4^{-x} = \tfrac{4}{3}, \qquad \mathbb{E}\Lambda_s = \sum_{i=1}^{s} \mathbb{E}\Delta_i \le \tfrac{4}{3}s.$$

By Markov's inequality $\mathbb{P}(\Lambda_s \ge 4s+1) \le \frac{4s/3}{4s+1} < \frac{1}{3}$, so $\mathbb{P}(\Lambda_s \le 4s) \ge \frac{2}{3}$.

Step 6 (one sample of density r). Let $W_* = W_1 \cup \cdots \cup W_{4s}$ (all $4s$ sets, including any that the procedure does not use). On the event $\{\Lambda_s \le 4s\}$ all successful draws are among $W_1, \dots, W_{4s}$,

so $\overline{W}_s \subseteq W_*$ and, by Step 4, W_* contains $\mathcal{U}(P)$ for some path P of $\mathcal{T}$. This has probability at least $\frac{2}{3}$. Each element of X lies in W_* independently with probability

$$r_* = 1 - (1-\rho)^{4s} \le 4s\rho = 16\mathrm{e}^{50}\sigma s \le 16\mathrm{e}^{50}\sigma(1+\log_2 \ell_k) = r,$$

using Bernoulli's inequality and $s \le 1 + \log_2 \ell_k$. So $W_* \sim \mu_{r_*}$ with $r_* \le r$. The event "W contains $\mathcal{U}(P)$ for some path P" is increasing in W, and μ_{r_*} and μ_r can be coupled so that the first set is contained in the second (take independent uniform variables U_e, $e \in X$, and the sets $\{e : U_e \le r_*\} \subseteq \{e : U_e \le r\}$). Hence the probability under μ_r is also at least $\frac{2}{3}$. □

4 Application to graph containment

It remains to construct the tree required by Theorem 3.1. The construction is the normalized-containment maximization and conditional-extension argument of Tran [16, Section 3, Lemma 3.1], with constants suited to the factor-16 separation of capacities; it is proved in full here.

Standing notation. Fix $n \ge 2$ and H as in Theorem 1.1, and put

$$q_* = p_{\mathrm{E}}(n, H), \qquad \sigma = 128 q_*, \qquad X = \tbinom{[n]}{2}.$$

Here q_* is a number, always defined from the original graph H with all its subgraphs. By facts (a) and (b) of Section 1, $q_* > 0$ and the constraints defining p_{E} hold at $p = q_*$.

Edge-set convention. A graph without isolated vertices is determined by its edge set, and we identify the two. So $E(H)$ denotes both the edge set of H and the graph obtained from H by deleting its isolated vertices, and an *edge set* $F \subseteq E(H)$ is regarded as the graph formed by the edges of F and their endpoints. That graph has no isolated vertices and is a subgraph of H in the sense of Section 1, so it is one of the graphs constrained in the definition of $p_{\mathrm{E}}(n, H)$. These are the only constraints of p_{E} that the proof uses; the constraints coming from subgraphs with isolated vertices are never needed. A subgraph of K_n isomorphic to such an F again has no isolated vertices and is determined by its edge set, so a *copy* of F in K_n may be regarded as a subset of X, and counting copies as subgraphs or as edge sets gives the same number. For edge sets F and I let

$$M(F) = N(K_n, F), \qquad M(I, F) = N(I, F),$$

the numbers of copies of F in K_n and in I; equivalently, $M(I, F)$ is the number of subsets of I that are isomorphic to F as graphs. These count copies, not vertex embeddings. For the empty edge set, $M(\varnothing) = M(I, \varnothing) = 1$. We use four facts.

(F1) *A graph on $[n]$ contains a copy of H if and only if its edge set contains a copy of $E(H)$.* For "if": the copy of $E(H)$ uses $v(H)$ minus the number of isolated vertices of H many vertices; since $n \ge v(H)$, enough further vertices of $[n]$ remain to serve as the isolated vertices, and copies need not be induced. "Only if" is clear.

(F2) *For every nonempty $F \subseteq E(H)$, $M(F) q_*^{|F|} \ge \frac{1}{2}$.* By the convention above the graph of F is a subgraph of H, so its constraint holds at q_*: $\mathbb{E}N(G(n, q_*), F) \ge \frac{1}{2}$. By (1.1), with $N(K_n, F) = M(F)$ and $e(F) = |F|$, the left side is $M(F) q_*^{|F|}$.

(F3) *If $F \subseteq F' \subseteq E(H)$, every copy S_0 of F in K_n is contained in a copy of F' in K_n.* Take an isomorphism from F to S_0, a bijection ϕ between their vertex sets. Since F' has at most $v(H) \le n$ vertices, ϕ extends to an injection of the vertex set of F' into $[n]$; the image of F' is a copy of F' containing S_0. In particular (take $F = \varnothing$) copies of F' exist.

(F4) *Copy ratio.* For an edge set $I \subseteq E(H)$ let $\mathbf{\Gamma}_I$ be a uniformly random copy of I in K_n. If $F \subseteq I$ and F_0 is any fixed copy of F in K_n, then

$$\mathbb{P}(F_0 \subseteq \mathbf{\Gamma}_I) = \frac{M(I,F)}{M(F)}. \tag{4.1}$$

Proof of (F4). A permutation π of $[n]$ acts on X and maps copies of an edge set to copies of the same edge set, so $\pi(\mathbf{\Gamma}_I)$ is again uniform on the copies of I. Any two copies F_0, F_0' of F satisfy $F_0' = \pi(F_0)$ for some π (extend an isomorphism between them to a permutation of $[n]$). Hence $\mathbb{P}(F_0' \subseteq \mathbf{\Gamma}_I) = \mathbb{P}(F_0 \subseteq \pi^{-1}(\mathbf{\Gamma}_I)) = \mathbb{P}(F_0 \subseteq \mathbf{\Gamma}_I)$: the probability is the same number θ for all copies. Summing over the $M(F)$ copies of F counts the copies of F contained in $\mathbf{\Gamma}_I$; there are exactly $M(I,F)$ of them because $\mathbf{\Gamma}_I$ is isomorphic to I. So $M(F)\theta = M(I,F)$. □

Lemma 4.1 (Graph hierarchy)**.** *There is a disjoint probability tree on $X = \binom{[n]}{2}$ whose path unions are copies of H with its isolated vertices removed, such that every level is σ-spread and its label sizes $\ell_1, \dots, \ell_k$ satisfy*

$$1 \le \ell_1, \qquad \ell_{i+1} \ge 16\ell_i, \qquad \ell_k \le e(H).$$

Here $k \ge 1$, and every label of level i has exactly ℓ_i elements, so level i has capacity ℓ_i. *Use:* this is the input to Theorem 3.1 in the proof of Theorem 1.1.

Proof. Plan. For an edge set I, choose $S \subseteq I$ maximizing a normalized containment probability. The constraints (F2) force $|S| < |I|/17$ (Step 1). Iterating from $I = E(H)$ gives a chain (Step 2), the tree records all ways to extend copies along the chain (Step 3), and maximality gives spread (Step 5).

Step 1 (the maximizer is small). For a nonempty edge set $I \subseteq E(H)$ and $S \subseteq I$ let

$$R_\sigma(I,S) = \sigma^{-|S|}\,\mathbb{P}(S_0 \subseteq \mathbf{\Gamma}_I) = \sigma^{-|S|}\,\frac{M(I,S)}{M(S)}, \tag{4.2}$$

where S_0 is any fixed copy of S; by (F4) this depends only on the isomorphism type of S. Choose $S \subseteq I$ maximizing $R_\sigma(I,S)$, breaking ties by a fixed rule. Since $R_\sigma(I,\varnothing) = 1$, the maximum is at least 1. We claim that $|S| < |I|/17$. This is clear if $S = \varnothing$. Otherwise write $h_I = |I|$ and $\kappa = |S| \ge 1$. By (F2), $1/M(S) \le 2q_*^\kappa$, and $M(I,S) \le \binom{h_I}{\kappa}$ since copies of S in I are κ-element subsets of I. Hence

$$R_\sigma(I,S) \le 2\binom{h_I}{\kappa}(q_*/\sigma)^\kappa = 2\binom{h_I}{\kappa}128^{-\kappa}.$$

If $\kappa \ge h_I/17$, the bound $\binom{h_I}{\kappa} \le (\mathrm{e}h_I/\kappa)^\kappa \le (17\mathrm{e})^\kappa$ gives

$$R_\sigma(I,S) \le 2(17\mathrm{e}/128)^\kappa \le 2\cdot\frac{17\mathrm{e}}{128} < 0.73 < 1,$$

contradicting $R_\sigma(I,S) \ge 1$.

Step 2 (the chain). Start with $I = E(H)$ and repeatedly replace I by its chosen maximizer until the empty set is reached; sizes strictly decrease, so this terminates. Writing the sequence in reverse order,

$$\varnothing = H_0 \subset H_1 \subset \cdots \subset H_k = E(H), \qquad |H_{i-1}| < |H_i|/17,$$

where H_{i-1} maximizes $R_\sigma(H_i, \cdot)$ over subsets of H_i, and $k \geq 1$ because $E(H) \neq \varnothing$. Put

$$\ell_i = |H_i| - |H_{i-1}| > 0.$$

Then $\ell_1 \geq 1$, $\ell_k \leq |E(H)| = e(H)$, and $\ell_{i+1} > 17|H_i| - |H_i| = 16|H_i| \geq 16\ell_i$.

Step 3 (the tree). The root is the empty edge set (the unique copy of H_0). A node at depth $i-1$ is a history of nested copies ending in a copy S_0 of H_{i-1}; distinct histories are distinct nodes. Its children are all copies I_0 of H_i in K_n with $S_0 \subseteq I_0$, with the uniform distribution, and the arc to I_0 is labelled $I_0 \setminus S_0$. By (F3) every node at depth $i - 1 < k$ has at least one child, so all leaves have depth k.

Step 4 (disjointness, path unions, label sizes). Along a path the copies are nested, $\varnothing = S^0 \subseteq S^1 \subseteq \cdots \subseteq S^k$, and the labels are the differences $S^i \setminus S^{i-1}$. They are pairwise disjoint, their union is S^k, a copy of $H_k = E(H)$, and $|S^i \setminus S^{i-1}| = |H_i| - |H_{i-1}| = \ell_i$.

Step 5 (every level is σ-spread). Fix a node at depth $i-1$ with current copy S_0 of H_{i-1}, and let A be the random label of the arc to a random child. The child is uniform among the copies of H_i containing S_0, that is, it has the law of $\mathbf{\Gamma}_{H_i}$ conditioned on $S_0 \subseteq \mathbf{\Gamma}_{H_i}$, an event of positive probability by (F3). Let $\varnothing \neq J \subseteq X$. If $J \cap S_0 \neq \varnothing$, then $\mathbb{P}(J \subseteq A) = 0$. Otherwise

$$\mathbb{P}(J \subseteq A) = \frac{\mathbb{P}(S_0 \cup J \subseteq \mathbf{\Gamma}_{H_i})}{\mathbb{P}(S_0 \subseteq \mathbf{\Gamma}_{H_i})}. \tag{4.3}$$

If the numerator is zero the bound $\mathbb{P}(J \subseteq A) \leq \sigma^{|J|}$ is trivial. If it is positive, $S_0 \cup J$ lies in some copy of H_i; pulling back along the isomorphism from H_i to that copy shows that $S_0 \cup J$ is a copy of some edge set $U \subseteq H_i$ with $|U| = |S_0| + |J|$. By (4.2) and (F4), $R_\sigma(H_i, U) = \sigma^{-|S_0|-|J|}\mathbb{P}(S_0 \cup J \subseteq \mathbf{\Gamma}_{H_i})$ and $R_\sigma(H_i, H_{i-1}) = \sigma^{-|S_0|}\mathbb{P}(S_0 \subseteq \mathbf{\Gamma}_{H_i})$. Maximality of H_{i-1} among subsets of H_i gives $R_\sigma(H_i, U) \leq R_\sigma(H_i, H_{i-1})$, that is,

$$\sigma^{-(|S_0|+|J|)}\mathbb{P}(S_0 \cup J \subseteq \mathbf{\Gamma}_{H_i}) \leq \sigma^{-|S_0|}\mathbb{P}(S_0 \subseteq \mathbf{\Gamma}_{H_i}).$$

With (4.3) this is $\mathbb{P}(J \subseteq A) \leq \sigma^{|J|}$. □

Proof of Theorem 1.1. Apply Theorem 3.1 to the tree of Lemma 4.1, with $\sigma = 128q_* > 0$ and capacities $\ell_1, \ldots, \ell_k$. It shows that a random subset of X of density $r' = \min\{1, 16\mathrm{e}^{50} \cdot 128q_*(1 + \log_2 \ell_k)\}$ contains a path union, that is, a copy of $E(H)$, with probability at least $\frac{2}{3}$. Since $16 \cdot 128 = 2048$ and $\ell_k \leq h = e(H)$,

$$r' \leq r := \min\{1,\ 2048\mathrm{e}^{50}q_*(1 + \log_2 h)\},$$

and by the monotone coupling of Step 6 in the proof of Theorem 3.1 the same holds at density r. The edge set of $G(n, r)$ has law μ_r, so by (F1), $\mathbb{P}(N(G(n, r), H) > 0) \geq \frac{2}{3} \geq \frac{1}{2}$, and $p_{\mathrm{c}}(n, H) \leq r$ by the definition of p_{c}. As $q_* = p_{\mathrm{E}}(n, H)$, this proves (1.2). Finally, $h \leq \binom{n}{2} \leq n^2$ and $\log_2 n \geq 1$ for $n \geq 2$ give $1 + \log_2 h \leq 1 + 2\log_2 n \leq 3\log_2 n$, and $3 \cdot 2048 = 6144$, which is the stated constant $6144\mathrm{e}^{50}$. □

5 Verification checklist

No step of this rewrite carries an "Unverified step" box: each step of the source was rechecked while writing and none raised a doubt. That is a report, not evidence; the list below says where a sceptical reader should look first.

1. **The one-pair computation** (2.10) (Lemma 2.1, Step 3). Check the cancellation $\rho^{-|A_b|}\rho^{|A_b\setminus A_c|} = \rho^{-|A_c\cap A_b|}$ and that independence needs f to ignore $W \cap A_b$. Both conclusions of Lemma 2.1 follow from it, (2.9) and (P4).

2. D_b **depends only on** $W \setminus A_v(b)$ (Lemma 3.2, Step 4). This is the only use of disjointness along paths in the probability estimate, and it is what allows (2.6) to be applied with $f = D_b$ in Step 5 although the same W is used at all nodes.

3. **The exponents in the potential recursion** (Lemma 3.2, Steps 5–6): $\mathrm{e}^{m_j} \cdot \mathrm{e}^{-9m_j}$ and $\mathrm{e}^{m_j} \cdot \mathrm{e}^{11m_j}$ in (3.6), the induction for (3.7), and $12 \cdot \frac{1}{15} - 8 \le -7$ from the factor-16 growth, giving failure probability below $\frac{1}{4}$.

4. **Spread of the reduced tree** (Lemma 3.2, Step 8). It uses that at fixed W the source still has law ν_v, that $T \subseteq A_v(\mathsf{c})$ by (P1), and that the normalization is $\gamma_v \ge 1 - \mathrm{e}^{-m_j}$.

5. **Iteration bookkeeping** (Theorem 3.1, Steps 1–4): the product bound (3.10), which keeps the spread below $a = 4\sigma$ in every round; the invariants (I1)–(I3), including why trimming zero-capacity levels preserves them; and completion after exactly s successes.

6. **From** s **successes to density** r (Theorem 3.1, Steps 5–6): the conditional bound (3.11) given the whole history, the geometric tail of the waiting times, the Markov bound for $4s$ attempts, and $1 - (1-\rho)^{4s} \le 4s\rho \le r$.

7. **The hierarchy** (Lemma 4.1): Step 1, where (F2) is the only use of the definition of p_{E} and $2 \cdot 17\mathrm{e}/128 < 1$ forces $|S| < |I|/17$; and Step 5, where maximality is applied to the isomorphism type U of $S_0 \cup J$ via (F4).

8. **Constants** (proof of Theorem 1.1): $16 \cdot 128 = 2048$, $\ell_k \le h$, and $1 + \log_2 h \le 3 \log_2 n$.

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